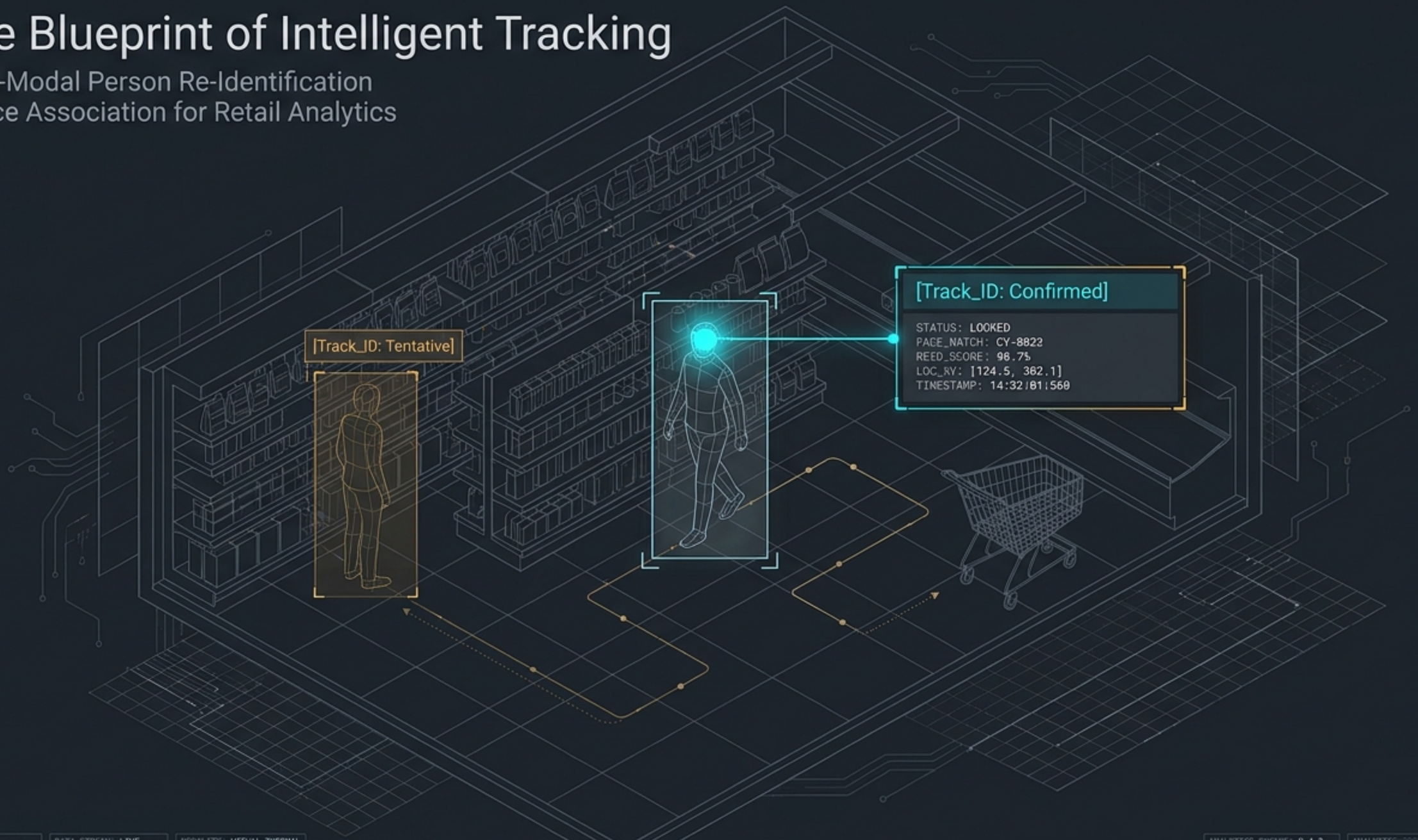
The Blueprint of Intelligent Tracking

Multi-Modal Person Re-Identification & Face Association for Retail Analytics



NotebookLM

Turning Fragmented Video into Persistent Global Identities



Harmonizing Two Modalities to Eliminate Blind Spots

	Body Re-Identification	Face Recognition
Strength	Operates at any distance and angle; requires no face visibility.	Extremely accurate; serves as absolute identity ground truth.
Weakness	Susceptible to failure from similar clothing, like store uniforms.	Demands close-up, well-lit, front-facing captures.
System Role	Primary continuous identification.	Authoritative override and retroactive correction.

Synthesis Insight: Faces act as the undeniable Ground Truth. When a face is matched, it retroactively corrects historical Body-based tracking data.

NotebookLM

The Engine Room: A Staged AI Model Architecture

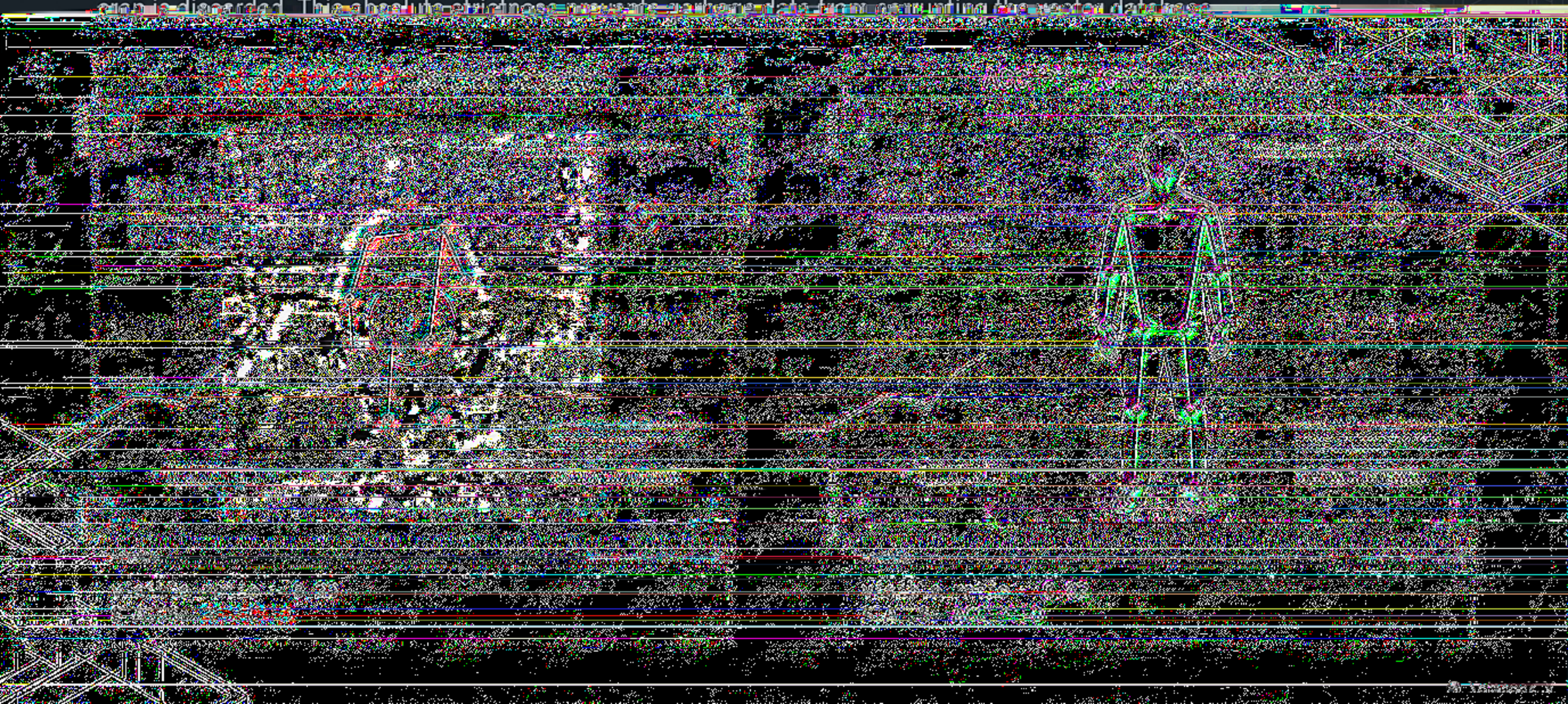
Model	Function	Architecture	Specs & VRAM	GPU
RT-DETR-L	[Person Detection]	[HGNet-v2 + Transformer]	NMS-free / 2-4 GB	cuda:0
YOLOv11s-Pose	[Quality Gate]	[YOLOv11 Small]	17 COCO Keypoints	cuda:1
BotSort	[Tracking]	[OSNet_x0_25]	Kalman Filter + IoU	CPU/Mixed
CLIP-ReID	[Body Vector]	[ViT-B/16]	1280-D Output	cuda:1
InsightFace Pack	[Face Vector]	[SCRFD + ResNet-50]	512-D / 1-2 GB	cuda:0

SYSTEM STATUS: Operating at real-time speeds on dual NVIDIA Tesla T4 GPUs (16 GB VRAM each).

NotebookLM

Filtering Noise with a Strict Skeletal Quality Gate

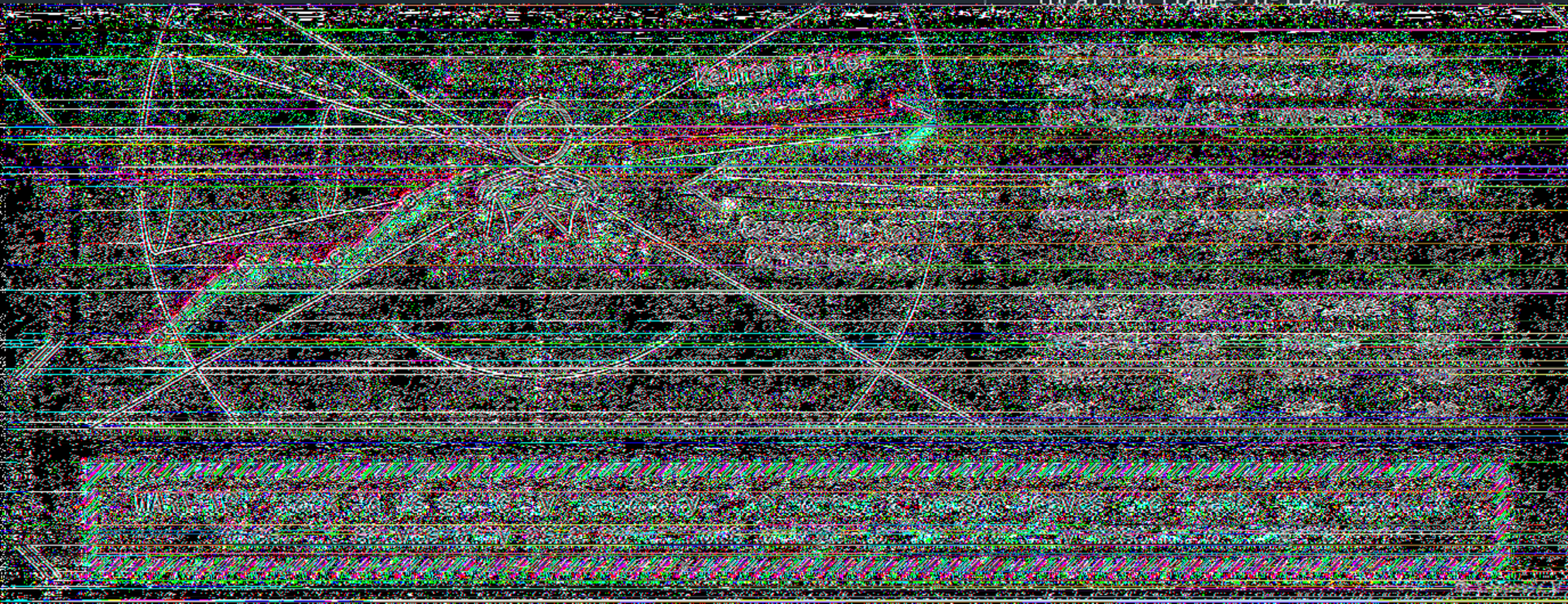
Before any mathematical embedding occurs, every detected crop must prove structural integrity. If visible_ratio < 0.5, the crop is discarded. This absolute strictness ensures only high-fidelity data from consenting dancers enters the vector database.



EDITORIAL COMMAND CENTER

Generating Temporary Continuity via BotSort

Kalman Filter:
Predicts expected spatial location frame-to-frame

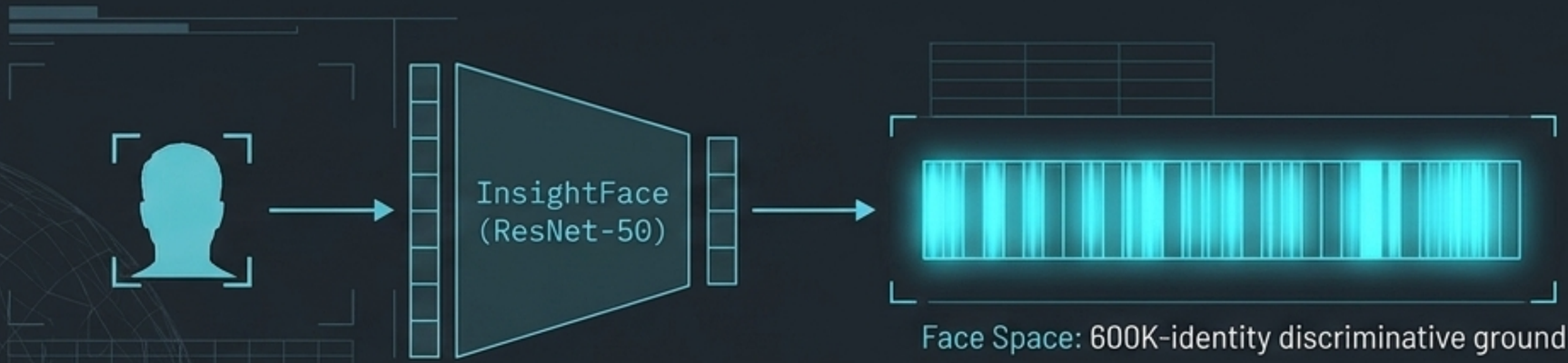


Translating Appearance into High-Dimensional Vectors

Body embeddings and Face embeddings are generated on parallel tracks. They exist in completely separate vector spaces and are never directly mathematically compared. They are linked solely by shared metadata.



Body Space: Captures clothing, texture, and silhouette. [1280-D]

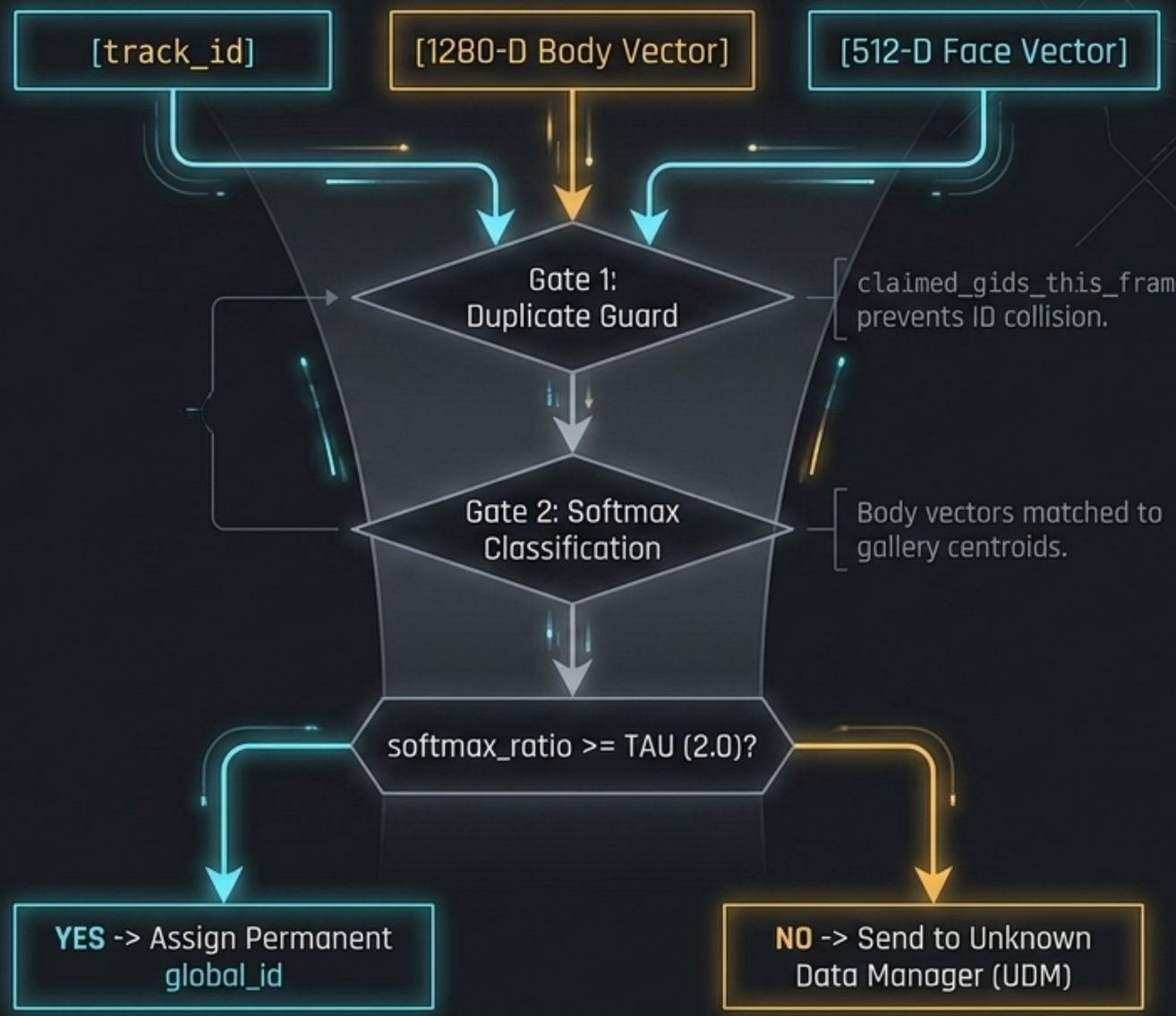


Face Space: 600K-identity discriminative ground truth. [512-D]

NotebookLM

The Identity Resolution Engine

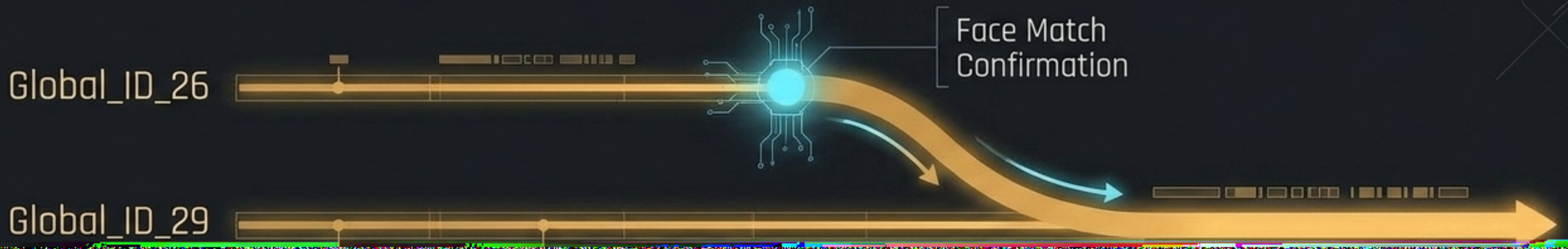
The system refuses to guess. If the temperature-scaled probability between the top match and the second-best match isn't at least double (TAU = 2.0), the system defers assignment rather than risking a false positive.



NotebookLM

The VIP Bypass: Retroactive Identity Merges

When ArcFace confirms that two separate global_ids share the exact same face, a cascading system merge executes instantly:



MANAGING AMBIGUITY WITH DBSCAN CLUSTERING

Unassigned crops wait in the UDM pool. Every 30 frames, DBSCAN evaluates the pool. If 5 or more crops form a tight visual cluster, the system takes action:



MINT

Create a brand new global ID.

MERGE

Absorb into an existing ID if the centroid is close enough.

INHERIT

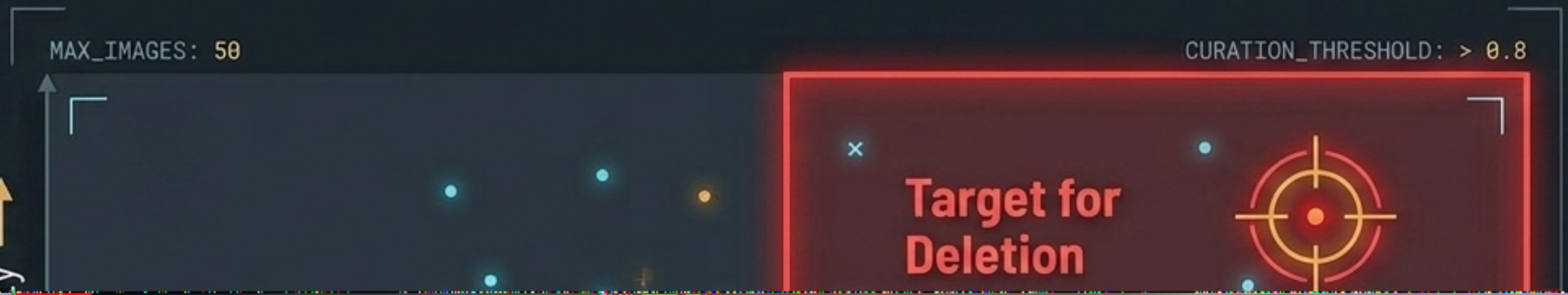
If a single face bypass exists within the cluster, the entire cluster instantly inherits that confirmed identity.

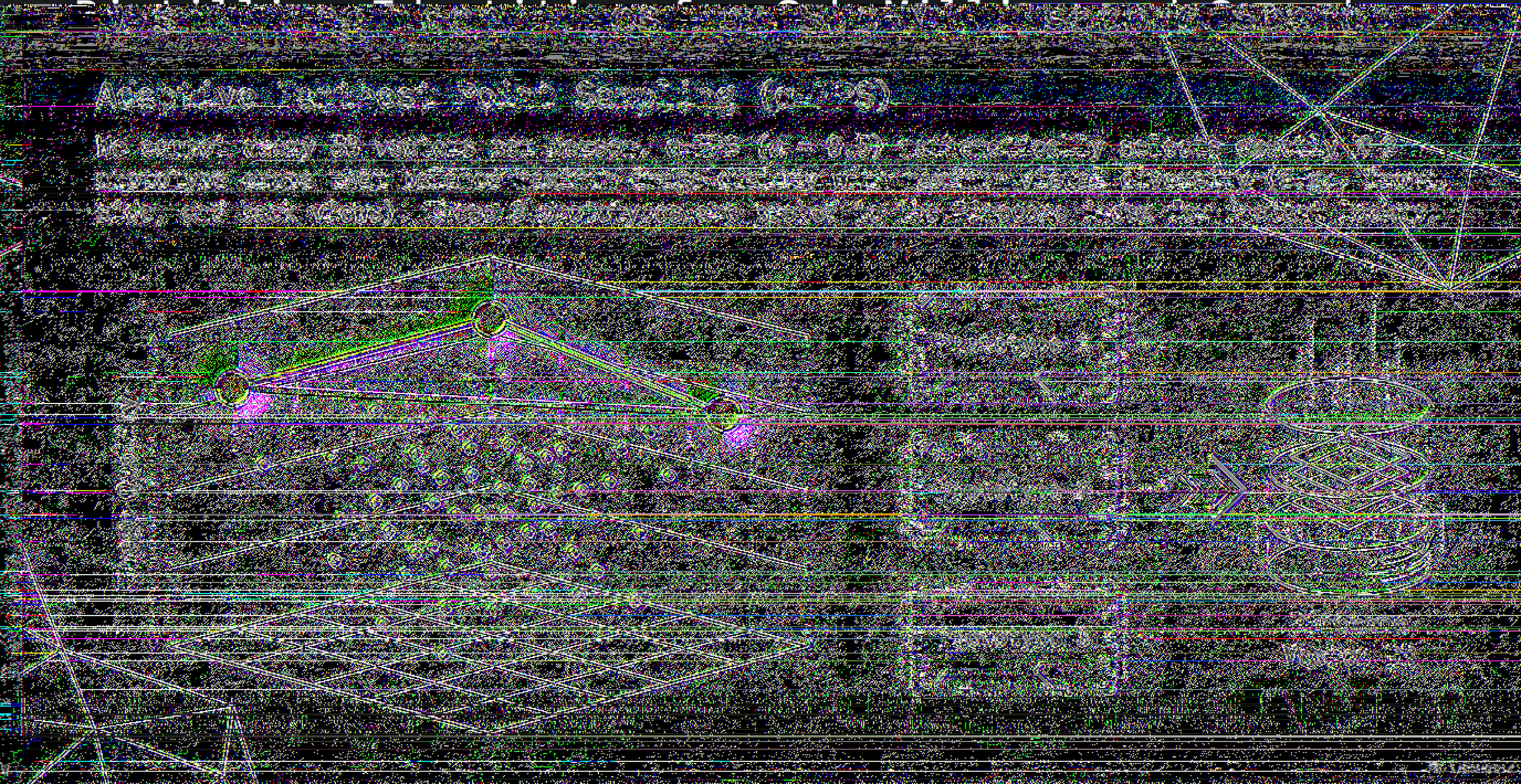
```
DBSCAN_MIN_SAMPLES: 5
DBSCAN_EPS: 0.15
POOL_EVAL_INTERVAL: 30 FRAMES
```

NotebookLM

The Self-Adaptive Identity Gallery

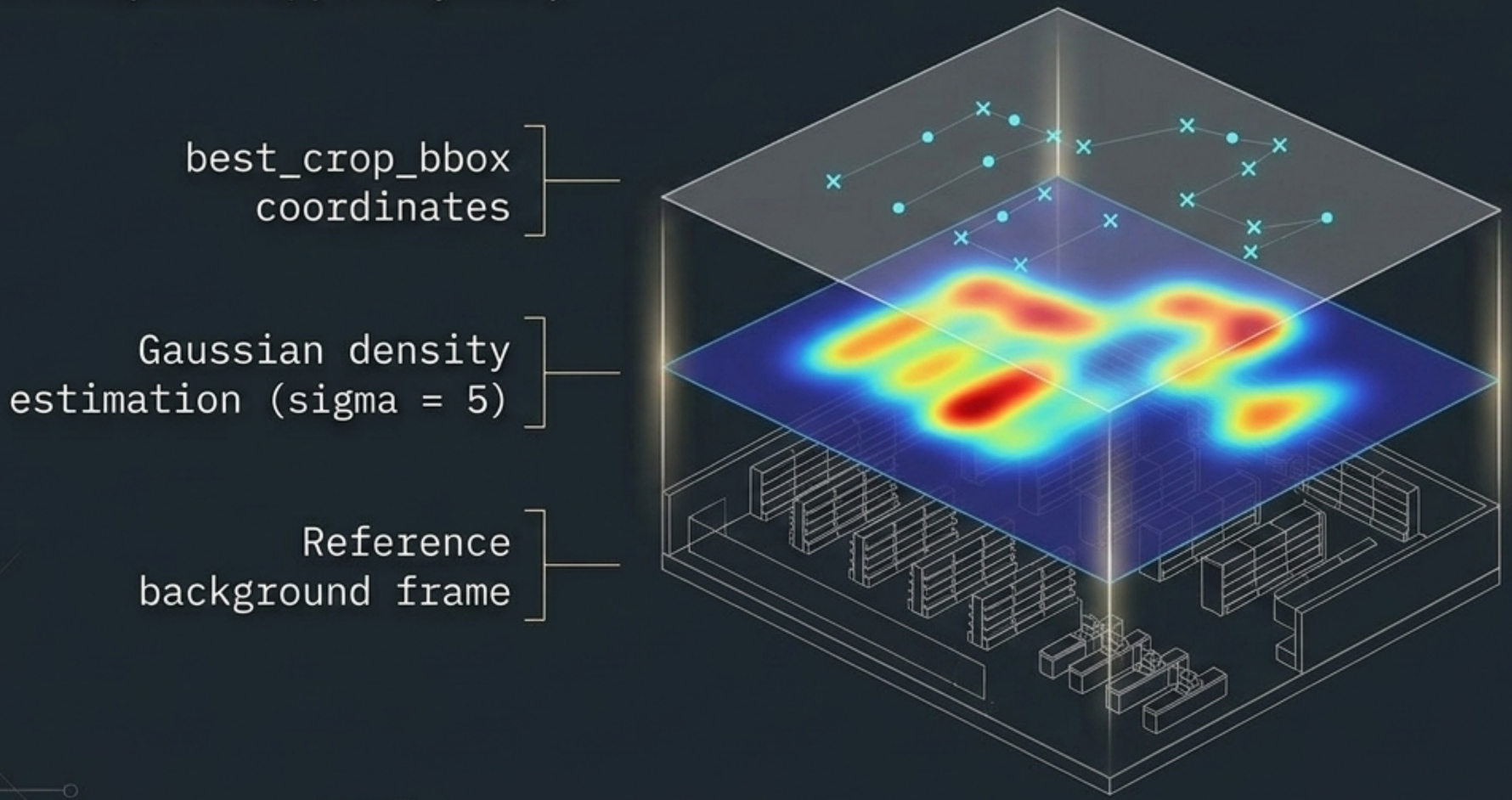
To manage VRAM limits, each person is capped at 50 saved crops (MAX_IMAGES). When the 51st crop arrives, the system self-curates by deleting the crop that is both highly redundant and highly uncertain.





Translating Mathematics into Physical Space

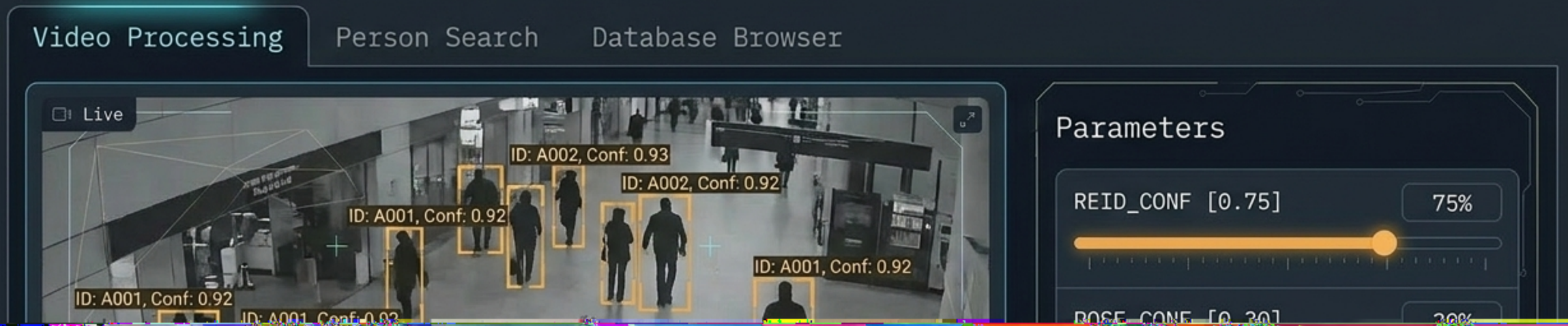
The system extracts absolute pixel coordinates, applies Gaussian density estimation, and maps it directly onto the reference background. High-dimensional vector matches resolve into a map of shopper trajectory.



NotebookLM

The Command Interface

The entire pipeline is exposed via a dynamic interface allowing real-time calibration.



Performance Validation: The Value of Refusing to Guess

By strictly enforcing the YOLO-Pose gate, the TAU=2.0 softmax ratio, and the Duplicate Guard, the pipeline ensures definitive assignment. Zero false positives in a production retail environment.

85.35%

Accuracy on Body Search

Brute force test across 1,317 crops

183 ms

Avg Query Speed

0.942

Avg Face Similarity Score

LOG // Out of 1,317 test queries, the system explicitly rejected 74 ambiguous crops. This is a feature, not a bug. The system strictly refuses to guess when unsure.

NotebookLM